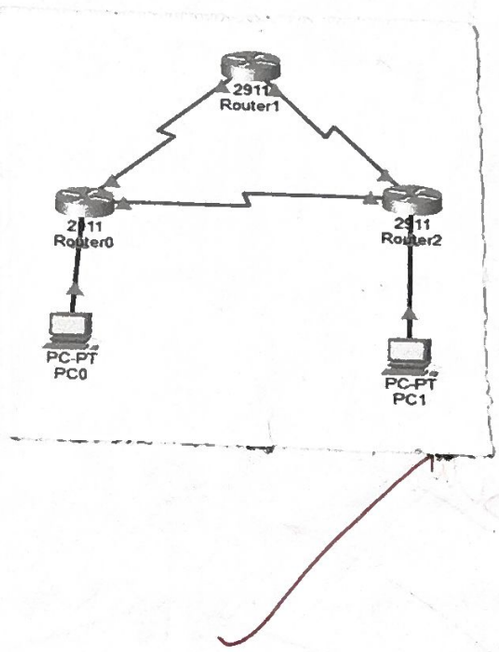
Exp No: 11 (a) STATIC ROUTING CONFIGURATION
29/09/26 USING CISCO PACKET TRACER

AIM:

To simulate static routing configuration using cisco packet tracer and verify the main and backup routes.

THEORY:

static routing is a routing method in which routes are manually configured in routers routing table. A static route specifies destination network, subnet mask, and next-hop address. Administrative distance (AD) used to slect a preferred route and configure a backup route. A route with lower AD over a route with higher AD.



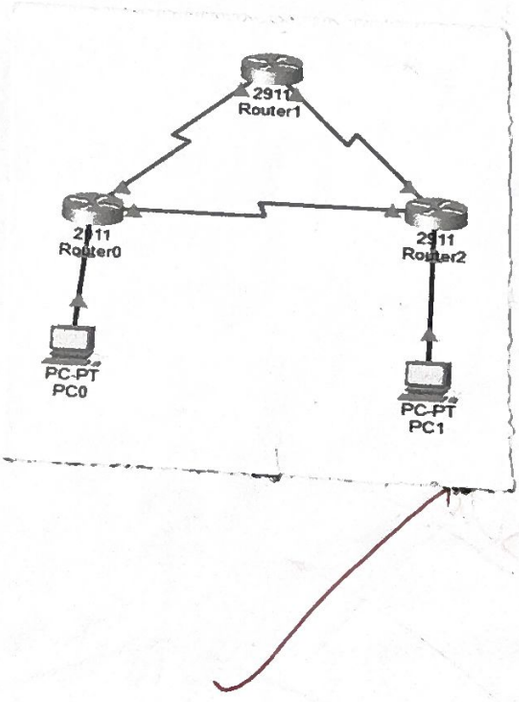
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IP ADRESSING TABLE :

Device	Interface	IP Address
PC0	Fast Ethernet0	10.0.0.2
Router 0	G0/0	10.0.0.1
Router 0	S0/3/0	20.0.0.2
Router 1	S0/3/1	20.0.0.2
Router 1	S0/3/0	50.0.0.1
Router 2	S0/3/1	50.0.0.2
Router 0	S0/3/1	40.0.0.1
Router 2	G0/0	40.0.0.2
Router 2	Fast Ethernet0	30.0.0.100
PC1		

Subnet masks:

/8 - 25.0.0.0

/32 - 255.255.255.255

PROCEDURE :

1. create the network topology using three routers and two pcs into cisco packet tracer.
2. Configure IP address of PCs and router interfaces.
3. Enable router interfaces using the no shut down command.
4. Configure preferred and backup routes using administrative values.
5. verify routes using show ip route static.
6. Use tracert to verify path followed by packets.

9. Disconnect main route between Router0 and Router1.

10. Verify that backup route is automatically selected.

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::290:2BFF:FE7E:8E4C
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.1.11
   Subnet Mask.....: 255.255.255.224
   Default Gateway.....: ::
                         192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0

C:\>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127
Reply from 192.168.2.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.11

Pinging 192.168.3.11 with 32 bytes of data:

Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time=14ms TTL=126
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 192.168.4.11

Pinging 192.168.4.11 with 32 bytes of data:

Reply from 192.168.4.11: bytes=32 time<1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126
Reply from 192.168.4.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

RESULT:

Thus, static routing was successfully simulated using Cisco packet tracer. The main and backup static routes were configured and verified. The route was automatically when primary route failed.